Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7/2	(a)	(i)		Concentration (of dye) (1) Do not allow: blue dye	1			1		1
		(ii)		Any 2 × (1) from: Length/size/diameter of chip (1) Temperature/ water bath at 20^{0}C / 20^{0}C (1) Volume of {solution/dye} (1) Same concentration (of dye) in each tube at start (1) Do not allow: How much potato / concentration / same time /colour / amount of dye	2			2		2
	(b)	(i)		(Concentration of dye/it) decreases (1)		1		1		1
		(ii)		{Movement/diffusion} (of dye) into {the cell/potato/chip/cylinder} (1) From high conc to low conc / down a concentration gradient (1) Do not allow: absorbed	2			2		2
	(c)			Decreases (at a decreasing rate) for 60 minutes (1) No change {between 60 and 80 minutes/ after 60 min/last 20 min} accept: last 10 min/stops changing at 60 min (1) If rate of diffusion was constant there would be a straight line sloping downwards (1) so disagree – judgement needed for (3) marks			3	3		3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			Start at 14 (1) Curve below the one plotted line <u>and</u> flattens at 7 before 60 minutes (1)			2	2	2	2
				Question 7 total	**5**	**1**	**5**	**11**	**2**	**11**